

IOWA STATE UNIVERSITY
ELECTRICAL AND COMPUTER ENGINEERING

EE 230
Temperature Sensor LED Design Lab Report

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Introduction

This lab was very interesting. We had the opportunity to create a circuit from scratch to fulfill different criteria. We designed the temperature sensor circuit where three different red, orange, and yellow LEDs dictate when temperatures are in different ranges. The circuit specifications are as follows:

As the temperature increases, the green LED will be lit for temperatures below 40°C and turn off past 40°C. As the temperature increases past 40°C, an orange LED will turn on, and finally, as the temperature passes 100°C, a red LED will turn on with the orange remaining on.

As the temperature decreases, the red LED stays on until the temperature falls below 80°C. The orange LED should stay on until the temperature falls below 30°C, and at this point the green LED will also turn on.

Design Process

We began by looking for ways to toggle the LEDs at given voltage values. These voltage values would directly translate to temperature values, with 0V representing 0°C and 1V representing 100°C. Through our research, we came across the comparator circuit that could be designed using the op-amps in our lab kits. To learn how we needed to configure the comparators, we began reading through Gary Tuttle's lecture slides on comparators.

Reading through the PDF on comparators, we found that they have a function of outputting voltages based on specified input threshold voltages. Basically, we could set a voltage trigger for which the LEDs would turn on and a voltage trigger for which the LEDs would turn off. This is exactly what we needed to fit our design requirements. Gary Tuttle's website also had a very convenient comparator spreadsheet solver. This allowed us to easily plug in values for our reference voltage, V_{L+} , V_{L-} , and play with our resistor values to find our specified V_{TL} and V_{TH} . For the functionality of the red and orange LEDs, this was pretty simple; we just had to use a non-inverting comparator and solve for our specified V_{TL} and V_{TH} . This is when we realized that our green LED would need to be a little more involved, as it had to turn off at a higher threshold than when we wanted it to turn on. So we had to use an inverting comparator. This allowed the LED to be on at a lower voltage and off at a higher voltage.

Design Equations

Although we used the Excel sheet to verify our calculations, it was important for us to know the equations we were using and how to more accurately pick resistor values and our upper and lower limit voltages. From the document, we were able to use these equations to help us narrow down our choices.

For the non-inverting comparators (red and orange LEDs) :

$$V_{TL} = V_{REF} \left(1 + \frac{R_a}{R_b} \right) - \left(\frac{R_a}{R_b} \right) V_{L+}$$

$$V_{TH} = V_{REF} \left(1 + \frac{R_a}{R_b} \right) - \left(\frac{R_a}{R_b} \right) V_{L-}$$

Where V_{TL} is the voltage threshold where we want our LED to turn on, and V_{TH} is the voltage threshold at which we want our LED to turn off.

For the inverting comparator (green LED):

$$V_{TL} = \frac{V_{REF} + \frac{R_a}{R_b} V_{L-}}{1 + \frac{R_a}{R_b}}$$

$$V_{TH} = \frac{V_{REF} + \frac{R_a}{R_b} V_{L+}}{1 + \frac{R_a}{R_b}}$$

Design Choices

As you may see, our circuit uses some unconventional design choices, in particular the fact that we use 3V as our V_{L+} and -3V as V_{L-} . This wasn't fully intentional, as we designed the comparators before fully understanding how they work. Optimally, we would've had it at 3V and 0V for V_{L+} and V_{L-} , respectively. However, since the LEDs will only let current flow in one direction, we don't really face any consequences to having a negative voltage on the output.

This unintentional design actually worked better in our favor as it allowed our R_A and R_B resistors for our comparators to all be very close to the resistor values we have in our lab kits, sparing us from having to use combinations of resistors in series and parallel to derive the required values.

Ultimately, we decided to have one voltage source acting as our V_{REF} at 0.9V for our red LED circuit and used a voltage divider to drop this value down to 0.35V for our orange and green LEDs. The red LED comparator needs a higher V_{REF} as it will not turn on until 1V and turns off again at 0.8V. Whereas the orange and green LED comparators will have a lower V_{REF} as their threshold voltages are very close to each other at the .3-.4V range.

Since the comparator output voltage was 3V, we placed 100Ω resistors in series with each LED to limit the current and prevent damage to the LEDs.

Calculations

Design Lab Calculations:

Comparator Equation: $\Delta V = V_{TH} - V_{TL}$

$$V_{TL} = V_{ref} \left(1 + \frac{R_a}{R_b}\right) - \left(\frac{R_a}{R_b}\right)V_L = \frac{\left(\frac{R_a}{R_b}\right)(V_L - V_{ref})}{1 + \frac{R_a}{R_b}}$$

$$V_{TH} = V_{ref} \left(1 + \frac{R_a}{R_b}\right) - \left(\frac{R_a}{R_b}\right)V_L = \frac{V_L - V_{ref}}{1 + \frac{R_a}{R_b}}$$

Final values:

Red LED: $\leftarrow$ Non-inverting	Green LED: $\leftarrow$ Inverting	Yellow LED: $\leftarrow$ Non-inverting
$R_a = 0.15k\Omega$ $V_{TL} = 1.02V$	$R_a = 0.1k\Omega$ $V_{TL} = 0.41V$	$R_a = 0.1k\Omega$ $V_{TL} = 0.41V$
$R_b = 4.7k\Omega$ $V_{TH} = 0.83V$	$R_b = 4.7k\Omega$ $V_{TH} = 0.28V$	$R_b = 4.7k\Omega$ $V_{TH} = 0.28V$
$V_{ref} = 0.9V$	$V_{ref} = 0.36V$	$V_{ref} = 0.36V$

Op-amp was powered by $\pm 3V$ DC

Computing ΔV_T :

Red LED:	Green LED:	Yellow LED:
$\Delta V_T = \frac{1.02V - 0.83V}{1 + \frac{0.15k\Omega}{4.7k\Omega}}$	$\Delta V_T = \frac{0.41V - 0.28V}{1 + \frac{0.1k\Omega}{4.7k\Omega}}$	$\Delta V_T = \frac{0.42V - 0.29V}{1 + \frac{0.1k\Omega}{4.7k\Omega}}$
$\Delta V_T = 0.1856$	$\Delta V_T = 0.127$	$\Delta V_T = 0.127$

ΔV_T is the band where current will flow; therefore, turning the LED on.

Computing V_{TL} :

Red LED:	Green LED:	Yellow LED:
$V_{TL} = 0.9V \left(1 + \frac{150\Omega}{4.7k\Omega}\right) - \left(\frac{150\Omega}{4.7k\Omega}\right)(-3V)$	$V_{TL} = 0.36V \left(1 + \frac{100\Omega}{4.7k\Omega}\right) - \left(\frac{100\Omega}{4.7k\Omega}\right)(-3V)$	$V_{TL} = 0.36V \left(1 + \frac{100\Omega}{4.7k\Omega}\right) - \left(\frac{100\Omega}{4.7k\Omega}\right)(-3V)$
$V_{TL} = 1.02V$	$V_{TL} = 0.41V$	$V_{TL} = 0.41V$

V_{TL} is the higher edge of the band where current will flow

Computing V_{TH} :

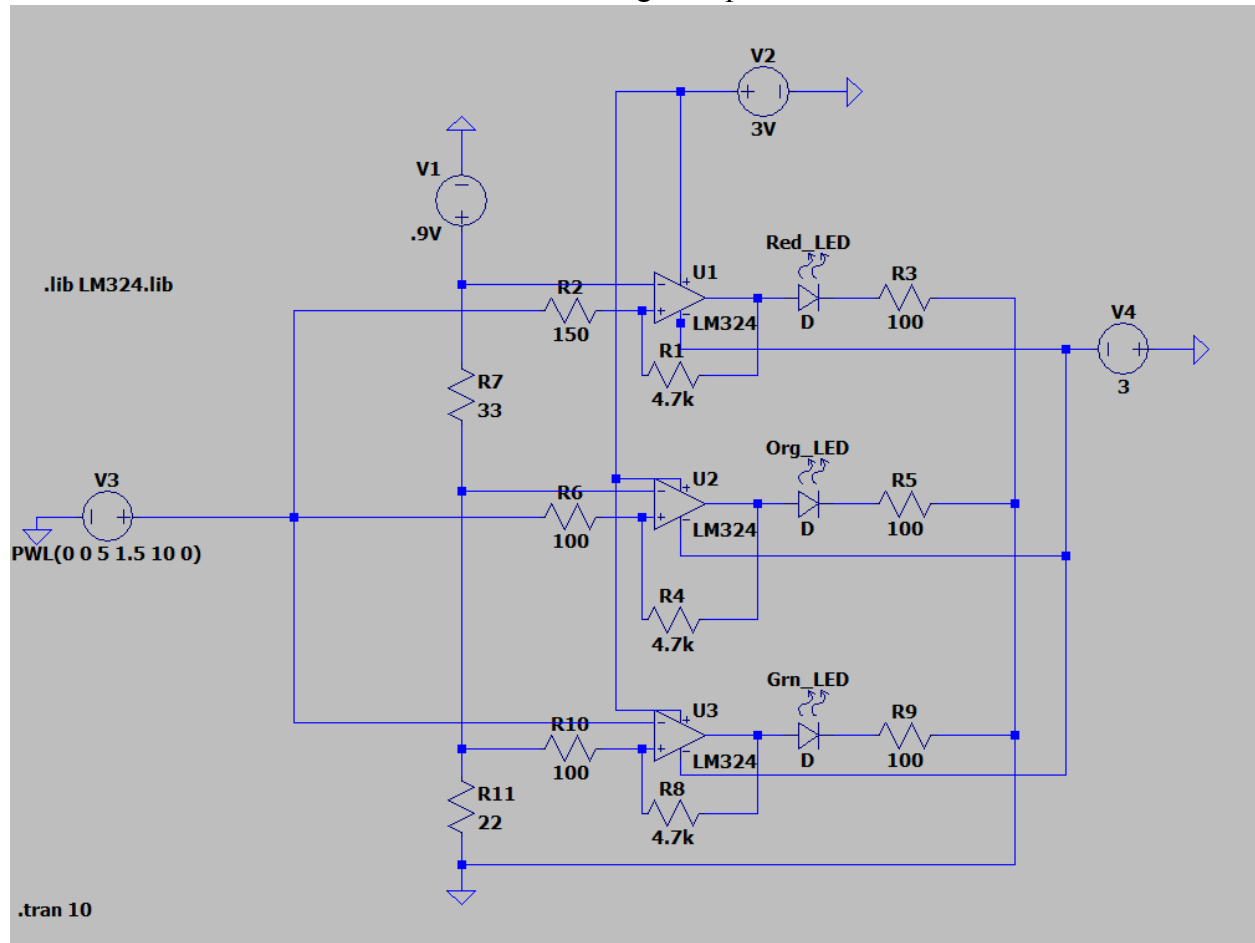
Red LED:	Green LED:	Yellow LED:
$V_{TH} = 0.9V \left(1 + \frac{150\Omega}{4.7k\Omega}\right) - \left(\frac{150\Omega}{4.7k\Omega}\right)(3V)$	$V_{TH} = 0.36V \left(1 + \frac{100\Omega}{4.7k\Omega}\right) - \left(\frac{100\Omega}{4.7k\Omega}\right)(3V)$	$V_{TH} = 0.36V \left(1 + \frac{100\Omega}{4.7k\Omega}\right) - \left(\frac{100\Omega}{4.7k\Omega}\right)(3V)$
$V_{TH} = 0.83V$	$V_{TH} = 0.28V$	$V_{TH} = 0.28V$

V_{TH} is the lower edge of the band where current will flow

This image shows our desired values for thresholds and a full list of our required resistors and voltage values for each comparator.

Schematic

After doing the calculations, we came up with the following circuit. We designed it in LTSpice and simulated it to ensure our LEDs were working as expected.

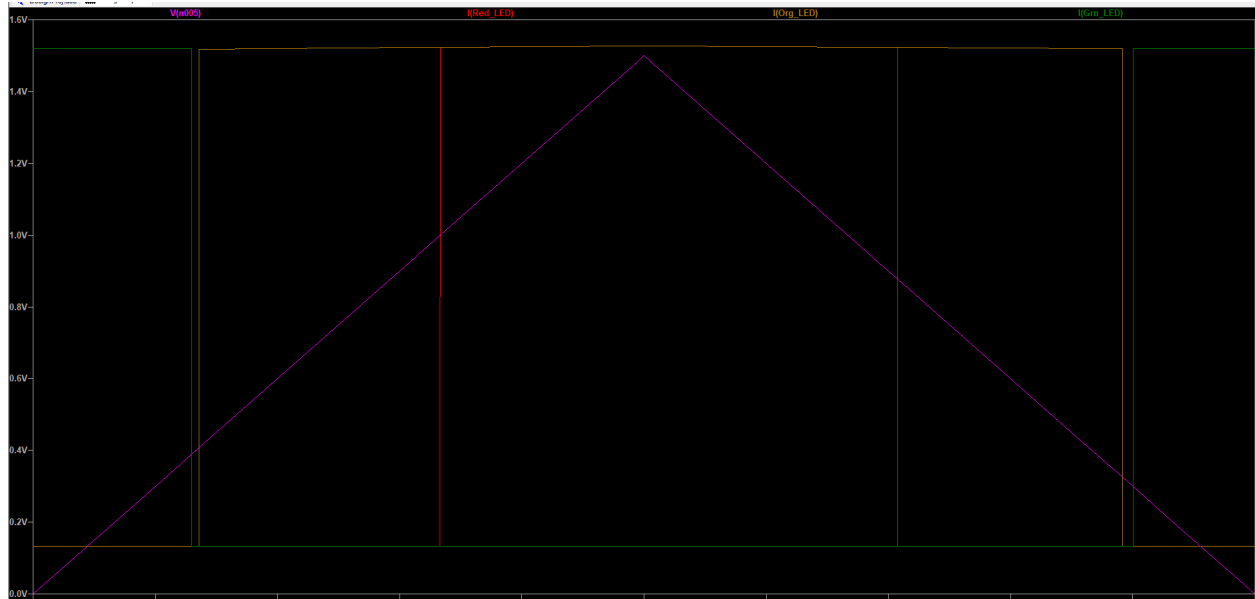


Simulation Output

Below is a screenshot of the simulation output for our simulated circuit.

The purple line is our input voltage going from 0V to 1.5V.

We can see our green LED stays on until the purple line hits $\sim 0.4\text{V}$, at which point the orange LED turns on. The orange LED stays on, and at $\sim 1\text{V}$, the red LED turns on. These LEDs both stay on until the purple voltage line drops back to $\sim 0.8\text{V}$, at which point the red LED turns off again. Then, finally, the orange LED turns off at around 0.3V , and the green LED turns back on.



Measurements and Testing

After simulating, we designed the circuit on our breadboard. Rather than using the analog output of the LM35 sensor, we tested functionality using a DC voltage supply. We slowly increased the voltage up to 1.5V and ensured the LEDs turned on and off as expected. At 1.5V , we slowly decreased the voltage back down to 0V . This was fairly simple to test as we didn't need to use the multimeter to confirm our results, we just had to watch the LEDs turn on and off.

Conclusion

We learned a lot from this lab. We ran into several challenges that allowed us to understand our circuit and logic better. We approached this project with a process, as we wanted to chunk this project out into three separate parts. We wanted to design three separate circuits, one for the red LED, one for the orange LED, and one for the yellow LED. In practice, this worked well. The challenge we faced was when we connected all of these circuits simultaneously. We discovered that the reference voltage acted as a voltage divider, which dropped our voltage to 0.27V instead of our desired voltage of 0.36V . We simply swapped out the resistors we were using to drop the voltage from 0.9V to 0.36V , and we were successful! This lab was a good exercise in perseverance and adaptability to the different situations we found ourselves in while designing this.

Bibliography

<https://gtuttle.net/electronics/topics/comparators.pdf>

<https://gtuttle.net/electronics/topics/comparators.xlsx>